

# MorSTORM user manual

## 1. Setting the environment

Please make sure Fiji/ImageJ and MATLAB are installed.

Please download the following packages:

A. MIJ<sup>1</sup> [MIJ: Running ImageJ and Fiji within Matlab - File Exchange - MATLAB Central](#)

B. ThunderSTORM<sup>2</sup> [ThunderSTORM](#)

C. MorphoLibJ<sup>3</sup> [MorphoLibJ](#)

## 2. Use the code

*2.1 Open CBSPcodeV2.m in the main folder and modify the key folders:*

```
1 %Enviroment setting:
2- javaaddpath 'C:\Program Files\MATLAB\R2021a\java\mij.jar'. %Path of mij.jar
3- javaaddpath 'C:\Users\zhang\Downloads\fiji-win64\Fiji.app\jars\ij-1.53f.jar'. %Path of Fiji
4
5 MIJ.start('C:\Users\zhang\Downloads\fiji-win64\Fiji.app');
6 Datafolder = 'C:\ADPtest\STMTEST2\'; %Path of the raw data folder
7- addpath(genpath('./AIM'))
8- N = dir(fullfile(Datafolder, '*.tif'));
9
```

Line 2: add the path where MIJ is placed. This should be in the MATLAB main folder.

Line 3: add the path where ij.jar is placed. This should be in the Fiji/ImageJ main folder.

Line 5: add the path of Fiji.app.

Line 6: folder where your data is stored. The code will process all .tif files in the folder.

*2.2 Setting ThunderSTORM and AIM parameters:*

The key parameters are identical to those in Fiji/ImageJ interference. However, here in MorSTORM, you can set the command directly in MATLAB and integrate with other MATLAB-based code.

```

10 % Change the parameters in the following lines:
11 % Line 36: Whether to remove certain frames (e.g. to avoid saturation) or not
12 % Line 37: Camera set-up
13 % Line 38: Gaussian fitting setting in ThunderSTORM
14 % Line 39: Post-fitting analysis in ThunderSTORM (Merge localisations to avoid replicates)
15 % Line 40: Post-fitting analysis in ThunderSTORM (Remove uncertainty-derived replicates)
16 % Line 41: Post-fitting analysis in ThunderSTORM (Remove noise by density filter)
17 % Line 63: Drift correction: AIM parameter
18 % Line 78: Final visualisation setting
19 % Line 98: Radius for localisation clustering
20

```

Line 36 is the command for removing certain frames:

```

36- | MIJ.run("Slice Remover", "first=1 last=200 increment=1");

```

This is derived from Image/Fiji's slice remover function. This can remove blurry/unstable/saturated frames.

Line 37 is the camera setting.

```

37- | MIJ.run("Camera setup", "offset=250 isemgain=false photons2adu=8 pixelsize=103.5");

```

Line 38 is the localisation identification command. This is the key command.

```

38- | MIJ.run("Run analysis", "filter=[Wavelet filter (B-Spline)] scale=2.0 order=3
detector=[Local maximum] connectivity=8-neighbourhood threshold=1.2*std(Wave.F1)
estimator=[PSF: Integrated Gaussian] sigma=1.6 fitradius=2 method=[Weighted Least squares]
full_image_fitting=false mfaenabled=false renderer=[Averaged shifted histograms]
magnification=10.0 dx=5.0 colorizez=false threed=false dzforce=false repaint=50");

```

The parameters are identical to the ImageJ ThunderSTORM interface. This command includes filter settings (type of filter and key parameters), approximate localisation of molecules (method, threshold, and neighbourhood setting), Gaussian fitting (Fitting parameters and method) and visualisation methods.

All parameters can be simply changed by replacing the texts with the desired command. For example, if you want to change the threshold:

```
threshold=1.2*std(Wave.F1)
```

Can be changed to

```
threshold=1.8*std(Wave.F1)
```

A ThunderSTORM parameter is also attached as a reference:

## Camera

Camera setup

## Image filtering

Filter: Wavelet filter (B-Spline) ?

B-Spline order: 3

B-Spline scale: 2.0

## Approximate localization of molecules

Method: Local maximum ?

Peak intensity threshold:  $1.2 * \text{std}(\text{Wave.F1})$ Connectivity: ☒ 8-neighbourhood☐ 4-neighbourhood

## Sub-pixel localization of molecules

Method: PSF: Integrated Gaussian ?

Fitting radius [px]: 3

Fitting method: Weighted Least squares

Initial sigma [px]: 1.6

Multi-emitter fitting analysis: ☐ enable

Maximum of molecules per fitting region: 5

Model selection threshold (p-value):  $1.0E-6$ ☐ Same intensity for all molecules☒ Limit intensity range [photons]: 500:2500

## Visualisation of the results

Method: Averaged shifted histograms ?

Magnification: 10.0

Update frequency [frames]: 50

3D: ☐Colorize z-stack: ☐

Z range (from:step:to) [nm]: -500:100:500

Lateral shifts: 2

Axial shifts: 2

Defaults

Preview

Ok

Cancel

Line 63 is the AIM parameters. While default parameters are recommended in the AIM paper, users can also easily modify other parameters following the AIM paper. [GitHub - YangLiuLab/AIM: Adaptive Intersection Maximization \(AIM\) is a high-speed drift correction algorithm for single molecule localization microscopy.](#)

## 2.2 Quantification:

The corrected and reconstructed images will then be converted into a binary format for quantification. By default, all morphological metrics will be quantified and stored in `mor_with_counts.csv`.

|    | A     | B     | C         | D           | E          | F          | G           | H           | I           | J           | K           | L           | M         | N         | O                 | P |
|----|-------|-------|-----------|-------------|------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-----------|-----------|-------------------|---|
| 1  | Label | Area  | Perimeter | Circularity | Centroid_X | Centroid_Y | Ellipse_Cer | Ellipse_Cer | Ellipse_Rac | Ellipse_Rac | Ellipse_Ori | Ellipse_Eto | MaxFeretD | MaxFeretD | LocalizationCount |   |
| 2  | 1     | 0.012 | 0.55      | 0.507       | 33.88      | 8.034      | 33.88       | 8.034       | 0.093       | 0.086       | -2.446      | 1.089       | 0.167     | 150.255   | 15                |   |
| 3  | 2     | 0.051 | 1.218     | 0.436       | 20.632     | 8.25       | 20.632      | 8.25        | 0.165       | 0.126       | -78.521     | 1.311       | 0.336     | 123.69    | 190               |   |
| 4  | 3     | 0.027 | 0.98      | 0.349       | 22.427     | 10.565     | 22.427      | 10.565      | 0.135       | 0.103       | 33.89       | 1.314       | 0.264     | 48.18     | 33                |   |
| 5  | 4     | 0.04  | 1.061     | 0.451       | 35.513     | 11.299     | 35.513      | 11.299      | 0.139       | 0.124       | -79.796     | 1.126       | 0.282     | 107.103   | 121               |   |
| 6  | 5     | 0.011 | 0.427     | 0.737       | 48.943     | 11.584     | 48.943      | 11.584      | 0.096       | 0.076       | 47.914      | 1.269       | 0.168     | 42.51     | 18                |   |
| 7  | 6     | 0.002 | 0.168     | 1.007       | 44.657     | 11.681     | 44.657      | 11.681      | 0.067       | 0.063       | -45         | 1.062       | 0.073     | 135       | 2                 |   |
| 8  | 7     | 0.047 | 1.108     | 0.481       | 36.851     | 12.07      | 36.851      | 12.07       | 0.147       | 0.131       | 66.216      | 1.124       | 0.304     | 72.181    | 185               |   |
| 9  | 8     | 0.052 | 1.191     | 0.464       | 47.196     | 12.377     | 47.196      | 12.377      | 0.147       | 0.143       | -15.678     | 1.029       | 0.305     | 170.218   | 487               |   |
| 10 | 9     | 0.031 | 1.013     | 0.378       | 32.724     | 13.523     | 32.724      | 13.523      | 0.138       | 0.106       | -87.489     | 1.309       | 0.272     | 107.745   | 52                |   |
| 11 | 10    | 0.005 | 0.288     | 0.781       | 48.716     | 13.689     | 48.716      | 13.689      | 0.074       | 0.071       | -59.9       | 1.038       | 0.102     | 135       | 6                 |   |
| 12 | 11    | 0.005 | 0.321     | 0.64        | 37.538     | 14.417     | 37.538      | 14.417      | 0.078       | 0.07        | 61.867      | 1.123       | 0.112     | 56.31     | 4                 |   |
| 13 | 12    | 0.014 | 0.57      | 0.552       | 23.304     | 14.934     | 23.304      | 14.934      | 0.096       | 0.089       | 18.754      | 1.073       | 0.168     | 47.49     | 17                |   |
| 14 | 13    | 0.045 | 1.092     | 0.471       | 37.795     | 18.33      | 37.795      | 18.33       | 0.143       | 0.13        | 82.195      | 1.101       | 0.271     | 133.452   | 217               |   |

## 2.3 Run the code:

Once all key parameters are settled, simply click run, and the code will automatically process all .tif in the folder. The processed data will be stored in the folder associated with the file. For example, if you have a file named X0Y0.tif, the code will create a folder called X0Y0P and store all associated data there. You will have:

- A. **loclist.csv**. The raw localisation list.
- B. **LocAIMOutput.csv**. The corrected localisation list.
- C. **loclist-protocol.txt**. ThunderSTORM parameters used for processing.
- D. **mor.csv**. Morphological metrics.
- E. **mor\_with\_counts.csv**. Morphological metrics with localisation number per molecule.
- F. **superres.tif**. Reconstructed super-res image.
- G. **filename\_AIM.mat**. Drift data from AIM.